

Equilibrium Bond–Length Distribution of a Harmonic Dumbbell

Why the mean bond length exceeds the rest length

PolyStokes — Brownian dynamics notes

July 1, 2026

Abstract

For two beads joined by a harmonic spring in three dimensions, the equilibrium probability density of the *scalar* bond length carries a geometric factor r^2 that is absent from the Boltzmann weight of the bond *vector*. As a result the mean bond length sits *above* the rest length r_0 , and for a soft spring ($k_{\text{bond}} r_0^2 \ll k_B T$) the deviation is large and set by the thermal width $\sigma = \sqrt{k_B T / k_{\text{bond}}}$ rather than by r_0 . This note derives the distribution, evaluates its mean exactly through a truncated–Gaussian moment recurrence, and connects the result to the purely–thermal dumbbell simulation.

1 Setup

Two beads at positions $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^3$ are joined by a harmonic bond that depends only on the length $r = |\mathbf{b}|$ of the bond vector $\mathbf{b} = \mathbf{x}_2 - \mathbf{x}_1$:

$$V(r) = \frac{1}{2} k_{\text{bond}} (r - r_0)^2. \quad (1)$$

In thermal equilibrium the Boltzmann distribution is over the full three–dimensional *vector* $\mathbf{b}$,

$$P(\mathbf{b}) \, d^3b \propto e^{-V(|\mathbf{b}|)/k_B T} \, d^3b. \quad (2)$$

As a function of the vector the weight is a Gaussian shell peaked where $|\mathbf{b}| = r_0$; by isotropy the mean vector vanishes, $\langle \mathbf{b} \rangle = \mathbf{0}$. This is the convention–free “zero on average” statement.

2 The length distribution and the r^2 factor

The distribution of the scalar length r is a different object. To obtain it we integrate (2) over all orientations at fixed length. In spherical coordinates the volume element factorises,

$$d^3b = \underbrace{r^2 \, dr}_{\text{radial}} \underbrace{\sin \theta \, d\theta \, d\phi}_{d\Omega}, \quad (3)$$

and since the integrand is independent of the angles, $\int d\Omega = 4\pi$ pulls out a constant, leaving

$$\boxed{P(r) \, dr \propto r^2 e^{-V(r)/k_B T} \, dr, \quad r \geq 0.} \quad (4)$$

The prefactor r^2 is the geometric (phase–space) Jacobian: it is the surface area $4\pi r^2$ of the sphere of radius r . There are simply more ways—more orientations, a larger spherical shell—to realise a

long bond than a short one. This factor, not any driving force, is what shifts the typical length above r_0 .

It is convenient to define the *thermal width*

$$\sigma \equiv \sqrt{\frac{k_B T}{k_{\text{bond}}}}, \quad (5)$$

so that $e^{-V/k_B T} = e^{-(r-r_0)^2/2\sigma^2}$.

3 Exact mean via truncated–Gaussian moments

The mean length is the ratio of the third and second moments of the truncated Gaussian,

$$\langle r \rangle = \frac{\int_0^\infty r^3 e^{-(r-r_0)^2/2\sigma^2} dr}{\int_0^\infty r^2 e^{-(r-r_0)^2/2\sigma^2} dr} = \frac{I_3}{I_2}, \quad I_n \equiv \int_0^\infty r^n e^{-(r-\mu)^2/2\sigma^2} dr, \quad (6)$$

with $\mu \equiv r_0$. Writing $(r - \mu) e^{-(r-\mu)^2/2\sigma^2} = -\sigma^2 \frac{d}{dr} e^{-(r-\mu)^2/2\sigma^2}$ and integrating by parts, the boundary terms vanish at both limits for $n \geq 2$, and one obtains the recurrence

$$\boxed{I_n = \mu I_{n-1} + (n-1) \sigma^2 I_{n-2}}. \quad (7)$$

The two seeds retain the effect of the lower cutoff $r \geq 0$ through the standard normal cumulative distribution Φ :

$$I_0 = \sigma \sqrt{2\pi} \Phi\left(\frac{\mu}{\sigma}\right), \quad I_1 = \sigma^2 e^{-\mu^2/2\sigma^2} + \mu I_0. \quad (8)$$

Applying (7) gives $I_2 = \mu I_1 + \sigma^2 I_0$ and $I_3 = \mu I_2 + 2\sigma^2 I_1$, hence

$$\langle r \rangle = \frac{\mu I_2 + 2\sigma^2 I_1}{\mu I_1 + \sigma^2 I_0}. \quad (9)$$

Numerical evaluation. For the simulation parameters $k_{\text{bond}} = 1$, $k_B T = 1$ (so $\sigma = 1$) and $r_0 = \mu = 0.2$:

$$I_0 = 1.452, \quad I_1 = 1.271, \quad I_2 = 1.706, \quad I_3 = 2.882, \quad (10)$$

$$\langle r \rangle = \frac{2.882}{1.706} = 1.6895, \quad \boxed{\langle r - r_0 \rangle = +1.49}. \quad (11)$$

This agrees with direct numerical integration of (4) and with the measured value $\langle r - r_0 \rangle = +1.51 \pm 0.04$ from the equilibrated frames of the thermal dumbbell run.

4 Physical interpretation: two regimes

The behaviour is controlled by the dimensionless ratio r_0/σ , equivalently by $k_{\text{bond}} r_0^2/k_B T$. The mode of (4) satisfies $\frac{d}{dr} [2 \ln r - \frac{1}{2} k_{\text{bond}} (r - r_0)^2/k_B T] = 0$, i.e.

$$\frac{2k_B T}{r} = k_{\text{bond}} (r - r_0), \quad (12)$$

a balance between the entropic outward “pressure” $2k_B T/r$ from the r^2 factor and the restoring spring force.

Entropic (soft) limit, $r_0 \ll \sigma$. Here the Gaussian is broad and nearly flat near the origin, so $P(r) \approx r^2 e^{-r^2/2\sigma^2}$ —exactly the Maxwell–Boltzmann speed distribution—with mean independent of r_0 ,

$$\langle r \rangle \longrightarrow 2\sqrt{\frac{2}{\pi}}\sigma \approx 1.596\sigma. \quad (13)$$

The present run is deep in this regime ($k_{\text{bond}}r_0^2 = 0.04 \ll 1$): the Maxwell mean 1.60 nudged up by the small $\mu = 0.2$ shift gives $\langle r \rangle = 1.69$.

Energetic (stiff) limit, $r_0 \gg \sigma$. The Gaussian is sharply peaked far from the origin, r^2 is essentially constant across the peak, and

$$\langle r \rangle \approx r_0 + \frac{2\sigma^2}{r_0} = r_0 + \frac{2k_B T}{k_{\text{bond}} r_0}, \quad (14)$$

so $\langle r - r_0 \rangle \rightarrow 0$ as $k_{\text{bond}} \rightarrow \infty$.

Dataset	$\langle r - r_0 \rangle$	$\langle r \rangle$
Simulation (equilibrated, $t \geq 1$)	$+1.513 \pm 0.040$	1.713
Simulation (all frames)	$+1.379 \pm 0.041$	1.579
Theory, $r^2 e^{-V/k_B T}$	+1.490	1.690

Table 1: Mean bond-length deviation from the rest length $r_0 = 0.2$ for $k_{\text{bond}} = k_B T = 1$. The equilibrated simulation matches the r^2 -weighted theory to within one standard error.

5 Summary

$$\langle r \rangle \approx \begin{cases} 2\sqrt{2/\pi}\sigma \approx 1.60\sqrt{k_B T/k_{\text{bond}}}, & r_0 \ll \sigma \quad (\text{entropic / Maxwell}), \\ r_0 + \frac{2k_B T}{k_{\text{bond}} r_0}, & r_0 \gg \sigma \quad (\text{energetic}). \end{cases} \quad (15)$$

The nonzero mean deviation $\langle r - r_0 \rangle \approx 1.5$ observed in the thermal dumbbell test is therefore *correct* equilibrium behaviour—an entirely entropic effect of the $4\pi r^2$ shell area—and not a symptom of a spurious force. The convention-free diagnostic for an unbiased integrator remains the mean bond *vector* $\langle \mathbf{b} \rangle$ and its per-sample increments, both of which vanish. To make the scalar deviation small one must enter the stiff regime, $k_{\text{bond}}r_0^2 \gg k_B T$.